Standardization that Enables Innovation: Building Deployment Confidence for Scalable Liquid Cooling

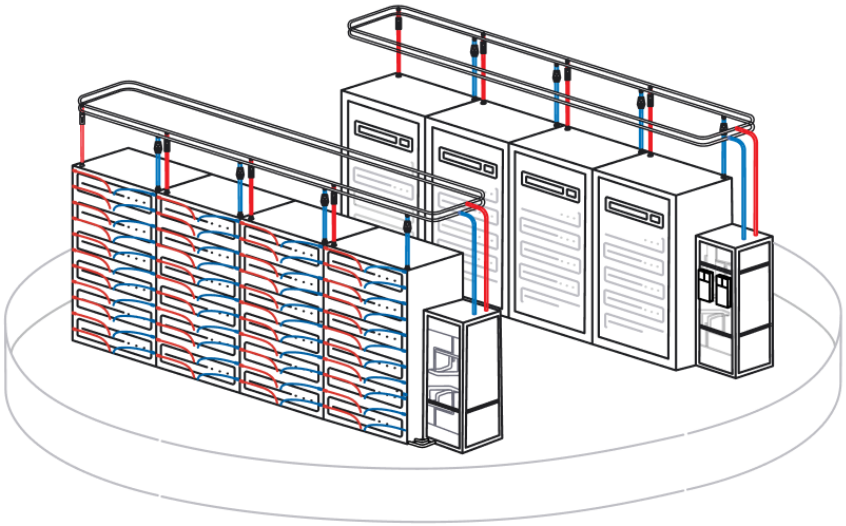
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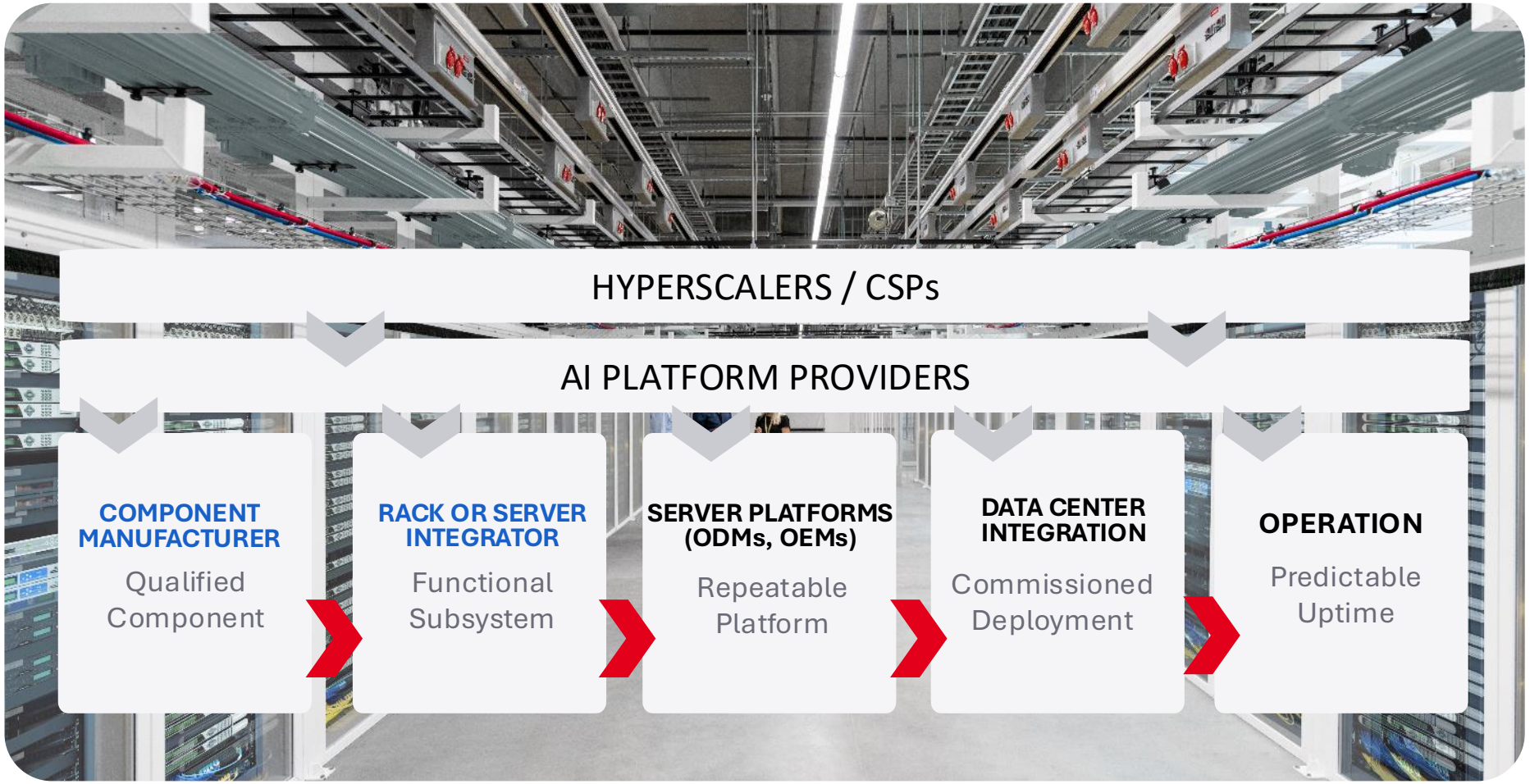
The challenge has moved

The question is no longer whether liquid cooling works. Liquid cooling technology is now proven.



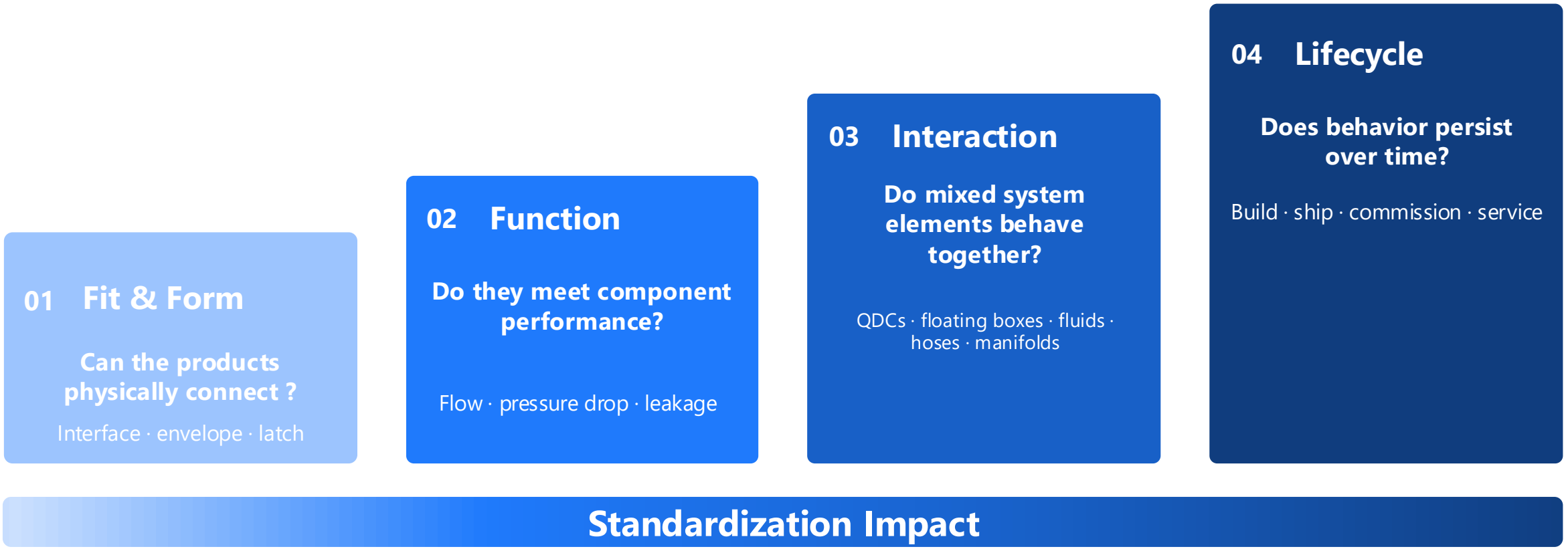
The challenge

is whether an ecosystem of suppliers, integrators, factories, operators, and service teams can repeatedly deliver the same reliable outcome.



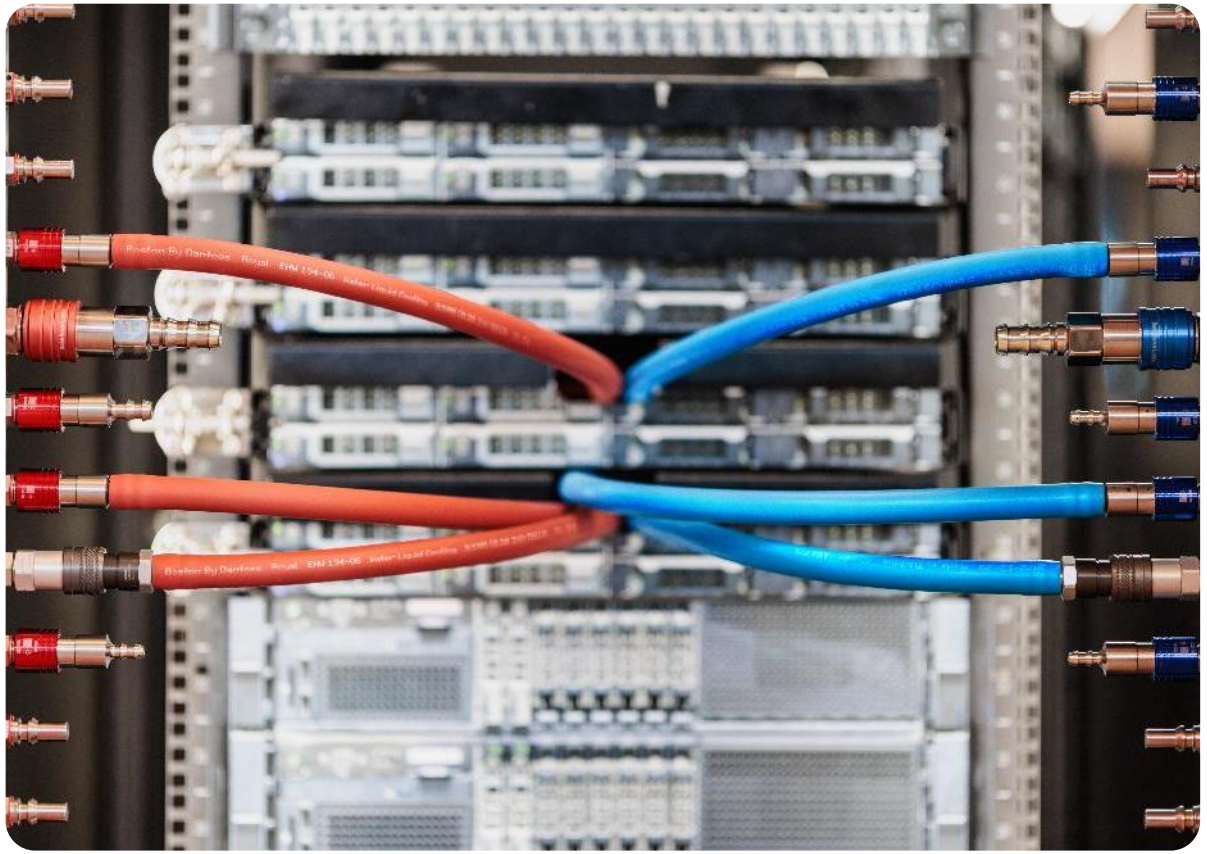
Every handoff introduces risk. Deployment at scale requires repeatability across the entire ecosystem.

How standardization helps to scale the deployment: Interchangeability is the entry point, not the destination



OCP Project Case Study

Universal Quick Disconnects (UQD) for liquid cooling



Standardized requirements

- Plug and Socket interface dimensions
- Engagement and panel-to-panel distance
- Pressure drop, fluid loss and cycling
- Mixed-supplier validation
- Marking, materials and test reporting

OCP Project Case Study

Universal Quick Disconnects (UQD) for liquid cooling

Lessons learned during interaction phase: How much mixed system elements behave together?

Characterize Floating functions:

- Pivoting reaction force
- Friction forces
- Assembly conditions
- Lifecycle exposure

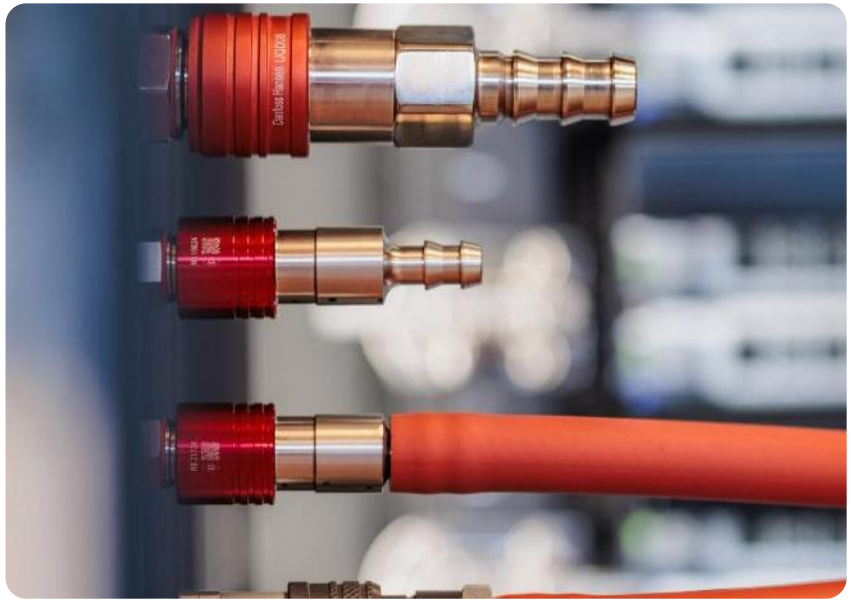


Key learning:
Consider standardization of an integrated floating function

OCP Project Case Study

Suppliers do not need identical internal solutions to deliver a common external promise.

Enabling innovation drives performance and reliability beyond the standard.



Standardize the desired outcome, not the design.



Standardize the desired outcome, not the design



Standardize

- Full interface and not only components
- System-relevant performance outcomes and lifecycle obligations
- Mixed-supplier interoperability evidence
- Relevant cleanliness expectations and acceptance criteria at lifecycle handoffs
- A shared definition of reliability for the specified duty cycle.

Enable innovation to drive performance and reliability

- Internal construction / architecture
- Integrated floating functions
- Manufacturing and inspection methods
- Flow-path optimization
- Diagnostics and reliability improvements beyond the minimum

Desired outcome : boundaries, measurable outcomes, acceptance evidence, and lifecycle obligations.



THE CHALLENGE IS CLEAR

Reliability must hold across every ecosystem handoff.

THE NEXT STEP

Define how every handoff is specified, qualified and accepted.

Best practices for scaling production and deployment

Standardize the complete interface

Connector performance depends on alignment, mounting tolerances, rack trajectory, and assembly conditions.

1

Publish the entire operating conditions

Capture every lifecycle condition—from integration and server-platform assembly through commissioning—and account for it from the outset.

2

Make cleanliness measurable at every handoff

Define the method, gate, evidence, owner and disposition.

3

Preserve an end-to-end evidence chain

Connect material, process, test, shipment, installation and service information.

4

Bring us the problem - not the solution

This enables genuine innovation by addressing the whole system.

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